

Omar Tawfik

Data Scientist | Statistical Modeling, Causal Inference & Machine Learning
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SUMMARY

Data Scientist with several years of experience building end-to-end statistical and machine learning systems, from raw data through feature engineering, modeling, validation, and deployment, and translating their outputs into consequential business decisions. Applies causal inference methods (DiD, Synthetic Control, PSM) as a genuine differentiator to isolate true effects from confounding factors, quantifying **\$31M** in financial exposure and shaping strategy that reduced projected costs by **\$9M**. Comfortable owning a project from raw data through production analytics and executive-facing recommendation.

PROFESSIONAL EXPERIENCE

Office of Labor Relations & Collective Bargaining (OLRCB) <i>Data Scientist</i>	Jan 2026 – Present <i>Washington, DC</i>
- Built an end-to-end causal inference platform (Python, Flask, SQL) combining Difference-in-Differences, Synthetic Control, and Propensity Score Matching via inverse-variance weighting to estimate policy and workforce effects, validating results through Monte Carlo simulation. - Quantified \$31M in potential settlement liability for a 26,000+ person workforce by building a compensation and back-pay prediction pipeline, applying feature engineering, regression, and cross-validated statistical modeling to payroll and demographic data. - Replaced single-point fiscal estimates with probabilistic exposure ranges (P5-P95) by engineering multi-year forecasting models across 5 negotiation strategies, combining time-series forecasting with sensitivity analysis to reduce projected fiscal exposure by approximately \$9M across 10+ negotiations. - Architected a centralized analytics and compliance framework with SQL-based data models, eliminating manual tracking gaps across 20+ compliance obligations.	
Booz Allen Hamilton Inc. <i>Data Scientist</i>	Jul 2023 – Nov 2024 <i>Washington, DC</i>
- Architected an end-to-end SQL and Azure ETL pipeline for CMS and VA government client engagements, replacing manual reporting workflows and improving data accuracy 50% and productivity 10x across 10+ biweekly deliverables. - Led a 6-person team standardizing KPI definitions across 4 business units and automating reporting with Python and Spark, delivering consistent Tableau dashboards to executive stakeholders.	
Karavan Imports LLC <i>Supply Chain Data Analyst</i>	Apr 2022 – Jul 2023 <i>Remote</i>
Developed Prophet and XGBoost demand-forecasting models that reduced inventory holding costs 35%; built Spark data pipelines processing 50M+ monthly records across 32 distributors and 5 wholesale partners.	

PROJECTS

Telematics Driver Risk Intelligence Platform — Independent
- Built an end-to-end telematics risk-scoring platform (Databricks, PySpark, Delta Lake) transforming 906K+ trip records and 250K+ GPS pings into 132 behavioral features across 1,200 drivers on a Bronze/Silver/Gold architecture with a documented synthetic data-generation process. - Trained and calibrated a HistGradientBoosting classifier under strict temporal validation, achieving 0.776 ROC-AUC and a 3.58x top-decile lift (35.8% of adverse events captured in 10% of drivers), with SHAP-based explainability and reason codes served via a FastAPI endpoint with MLflow tracking and PSI-based drift monitoring.
Medicare Healthcare Cost & Utilization Intelligence Platform — Independent, In Progress
Built and deployed XGBoost regression for Medicare spending estimation and an XGBoost classifier to identify high-cost providers, achieving 0.925 ROC-AUC and 0.677 PR-AUC using temporal holdout validation, on a Bronze-Silver-Gold pipeline (DuckDB, SQL, Python) ingesting public CMS Medicare claims-summary data with suppression handling and leakage-prevention controls.
DocMind: RAG Document Intelligence Pipeline
Designed and deployed a Python/Flask RAG pipeline integrating a third-party LLM API and Qdrant vector store; validated retrieval with a 10-question harness, achieving 100% hit-rate@6 and 0.95 MRR.

TECHNICAL SKILLS

Machine Learning & Statistics: Predictive Modeling, Classification, Regression, XGBoost, Feature Engineering, Model Validation & Evaluation, Temporal Validation, Cross-Validation, Time-Series Forecasting (Prophet), Experimental Design
Causal Inference & Econometrics: Difference-in-Differences (DiD), Synthetic Control, Propensity Score Matching (PSM), Monte Carlo Simulation, Sensitivity Analysis
AI & LLM Engineering: Retrieval-Augmented Generation (RAG), LLM APIs, Vector Databases, Prompt Engineering, Retrieval Evaluation (MRR, Hit Rate)
Programming & Data Engineering: Python, SQL, scikit-learn, Flask, Apache Spark, ETL Pipeline Development, Data Pipeline Architecture, Databricks, Snowflake
Cloud & Visualization: Azure, Tableau, Power BI, Qlik, Chart.js

RELEVANT COURSEWORK & TRAINING

- University of Maryland, MS Data Science: DATA601 Probability and Statistics; DATA602 Principles of Data Science; DATA603 Principles of Machine Learning; DATA604 Data Representation and Modeling; DATA605 Big Data Systems; DATA606 Algorithms for Data Science; DATA607 Communication in Data Science. - Booz Allen Hamilton internal training: Tech Excellence Program (Data Science Track), TechTank, MITx/IDSS Statistics and Data Science coursework (sponsored partnership).

EDUCATION

University of Maryland, Robert H. Smith School of Business <i>Master of Science, Data Science</i>	May 2026
University of Maryland <i>Bachelor of Science, Operations Management & Business Analytics</i>	May 2023